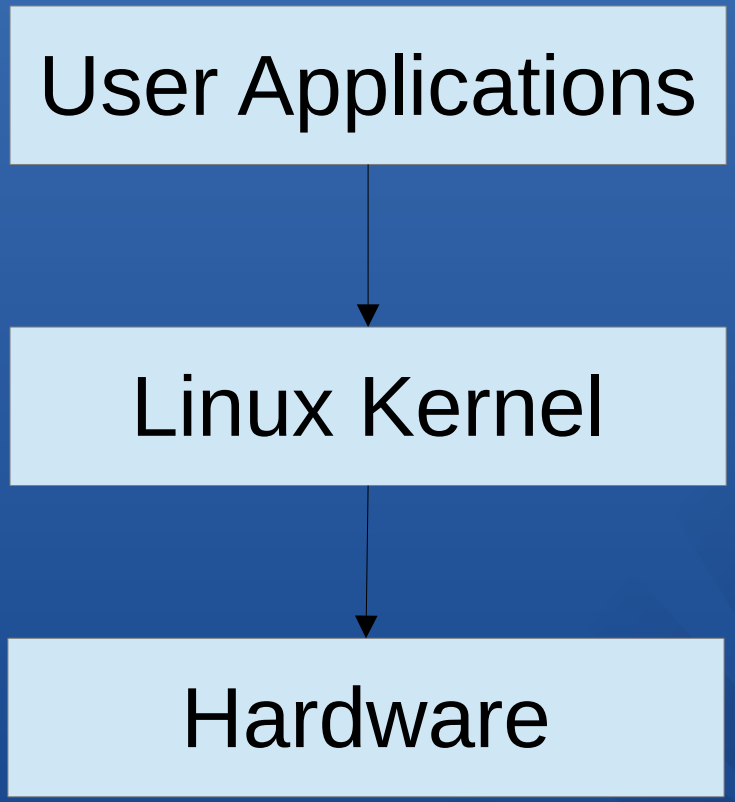
Linux Kernel and Booting

by Kowshic Shankar R

What is a Linux Kernel

- The Linux kernel manages the hardware and provides services to programs.
- Kernel's Services:
  - Process Management
  - Memory Management
  - Device Drivers
  - Filesystems



Bare Metal vs Linux (OS)

Bare Metal

- Single Program
- Direct Physical Register Access
- No Memory Protection
- Full Timing Control
- Fast Startup
- Limited Portability (Software)

Linux

- Many Processes
- Device Drivers
- Virtual Memory
- Scheduler Based Timing
- Booting Sequence
- Portability

Kernel vs Distro

Kernel (Engine)

- Low Level Software (Just the bare minimal kernal)
- Maintained by Linus Torvalds and other open source contributers
- Fully open source
- Kernal 1.0,2.0,6.8...

Distro (Car)

- Complete Operating System (Kernel + Shell + Package Manager + GUI + Applications etc...)
- Maintained by companies and communities.
- Many not be open source
- Ubuntu, Fedora, Arch Linux

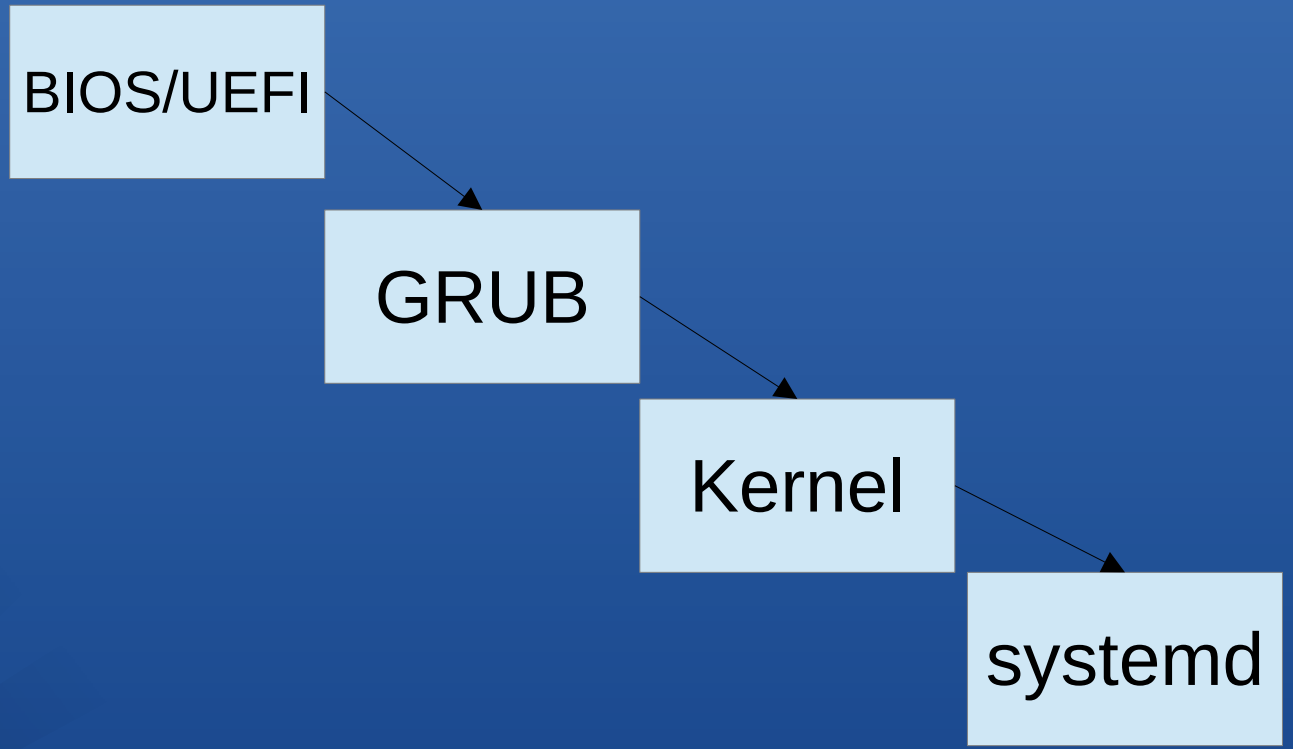
Monolithic Architecture

- Linux follows a Monolithic Architecture.
- Monolithic means that many core operating-system services run together inside one kernel space.
- Microkernel is the another Architecture. It is a minimal kernel with only the basic services. The Main difference is that the Device Drivers here run on User Space.
- Advantage of Monolithic Kernel is isolation of hardware completely from user space.
- Advantage of Microkernel is the faults occur in device drivers dosent affect the Kernel.
- Microkernel Example : MINIX, QNX

Loadable Kernel Modules

- A monolithic kernel can still be modular.
- A device driver may be compiled as a .ko file and loaded at runtime.
- Once loaded these binaries run in kernel space unlike normal binaries that run on user space.
- In a Linux shell run the command lsmod for listing the modules that are currently available.
- modinfo cmd gives the description, author and license of a module
- insmod cmd can insert a .ko module.
- rmmod cmd removes a module

Booting Sequence



Booting: BIOS

- The Motherboard contains a embedded firmware called BIOS or nowadays UEFI.
- It Excecutes Self tests for storage devices , RAM and CPU.
- In case of BIOS it search for a Master Boot Record (MBR) on the partitions of the storage devices. MBR Storage Devices has a limitation of 2TB.
- In case of UEFI it is done through EFI partitions.
- These MBR and UEFI loads the bootloader like GRUB, UBOOT, Windows Boot Manager etc.

Booting : GRUB

- GRUB (GRand Unified Bootloader) is a widely used bootloader that has a builtin filesystem that loads the kernel and initrd/initramfs.
- The initrd/initramfs is a temporary filesystem loaded on memory for the kernal until the actual filesystem is initiated by the kernel.
- GRUB is also used to select the kernels and pass the kernel parameters.
- Common kernel parameters are : quiet , nomodeset , splash

Booting : Kernel

- The Kernel and the initrd/initramfs can be found in the /boot partition.
- Vmlinuz is the kernal binary that is compressed using z-lib.
- The Kernal uncompress and progresses through a init system.
- There are various init systems like Sysvinit, upstart and systemd. Currently systemd is widely used.

Booting : systemd

- systemd is a software suite for system and service management on Linux
- Its main component is an init system used to bootstrap user space and manage user processes.
- It also provides various daemons and utilities, including device management, login management, network connection management, and event logging.
- systemctl is a command to introspect and control the state of the systemd system and service manager.